Interest, Taxes, and Phase Transitions in Debt Recycling

Extending the model with rates and tax shields and calibrating to AUS, DEU, CHE

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Methodology 1 — Theoretical

Goal Extend the base model from Aufiero et al. 2025 by adding mortgage and borrowing interest rates r_b and r_m and tax shields τ_b, τ_m.

State equity E_t and mortgage M_t updated quarterly

$$\begin{aligned} E_t &= E_{t-1} + \pi_t + I_{t-1} + H_{t-1} r_t - (1-\tau_b) r_b U_{t-1} + \tau_m r_m M_{t-1} \\ M_t &= (1+r_m) M_{t-1} - \pi_t - I_{t-1} \end{aligned}$$

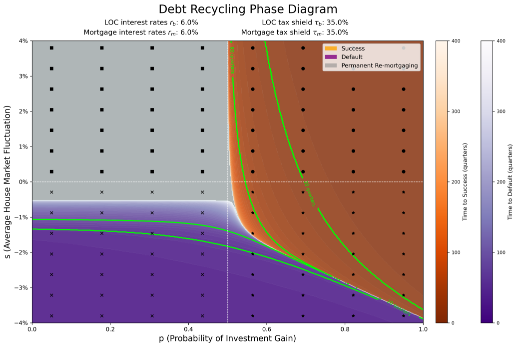
- $U_{t-1} = \ell E_{t-1}$ credit line against equity, $I_{t-1} = \sigma_{t-1} \mu U_{t-1}$ with $\Pr(\sigma = +1) = p$
- Average matrix $\bar{A}$ drives $\langle E_t \rangle, \langle M_t \rangle$ and sets regions and times

$$\begin{bmatrix} \langle E_t \rangle \\ \langle M_t \rangle \end{bmatrix} = \bar{A} \begin{bmatrix} \langle E_{t-1} \rangle \\ \langle M_{t-1} \rangle \end{bmatrix} + \begin{bmatrix} \langle \pi \rangle \\ -\langle \pi \rangle \end{bmatrix}, \quad \bar{A} = \begin{bmatrix} 1 + s + \ell\mu(2p-1) - \ell(1-\tau_b) r_b & s + \tau_m r_m \\ -\ell\mu(2p-1) & 1 + r_m \end{bmatrix}.$$

- Outcomes by first hit of the mean paths within 400 quarters success if M hits zero first, default if E hits zero first, otherwise permanent re-mortgaging

Results 1.1 — Phase Diagram

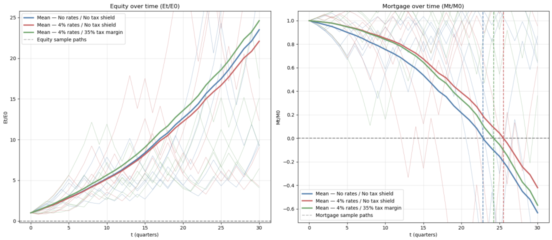
- Adding positive r_b and r_m shrinks success and slows repayment where success still occurs
- Tax shields τ_b, τ_m lower effective costs and can bring back part of the lost region and shorten times
- Phase diagrams show boundary shifts across (p, s) and Monte Carlo gives outcome times under the same rules
- Grey area means permanent re-mortgaging when neither boundary is hit within the horizon



Results 1.2 — Monte Carlo

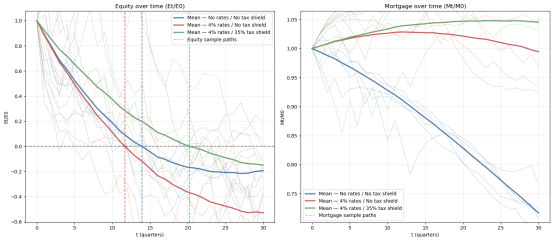
- Confirms the phase map story: no rates pay off fastest, rates without tax shields are slowest, tax shields sit in between
- With rates on, time to success increases; adding shields trims part of the delay
- With low p and negative s, shields mostly delay failure rather than turn it into success
- Shields help most when $p > 0.5$ and s is near zero or positive
- Same inputs as in the phase diagrams, so theory and simulation are directly comparable

Success scenario



Mean path bold; first hit of $M = 0$ marks payoff time

Default scenario



Mean path bold; first hit of $E = 0$ marks default time

Methodology 2 — Calibration

What we calibrate

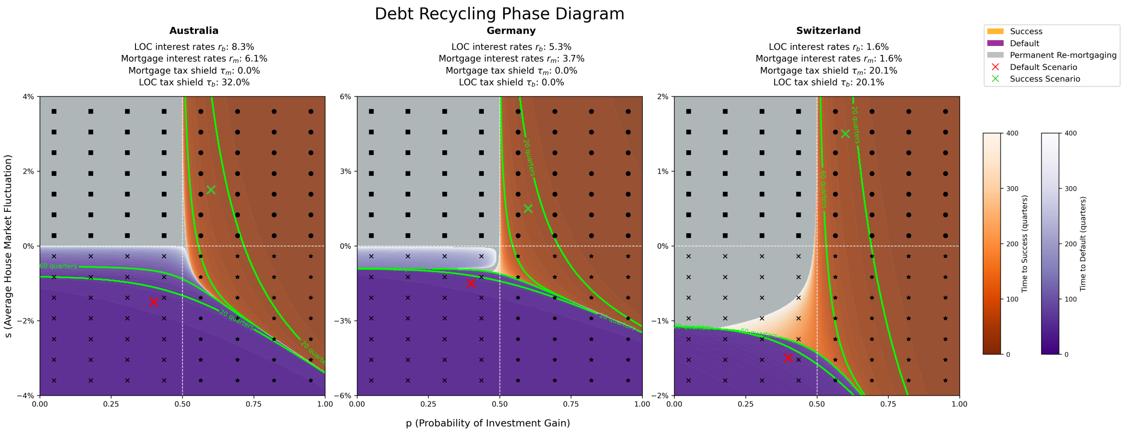
- Annual averages mid 2024 to mid 2025 for r_m, r_b, τ_m, τ_b, housing drift band s
- Two use cases per country self-owned and rental
- Marginal tax rates compute τ from country tax code functions and feed directly into the model

How we use it

- Insert calibrated inputs into $\bar{A}$ for phase diagrams
- Run Monte Carlo with the same parameters to get first passage times for success and default

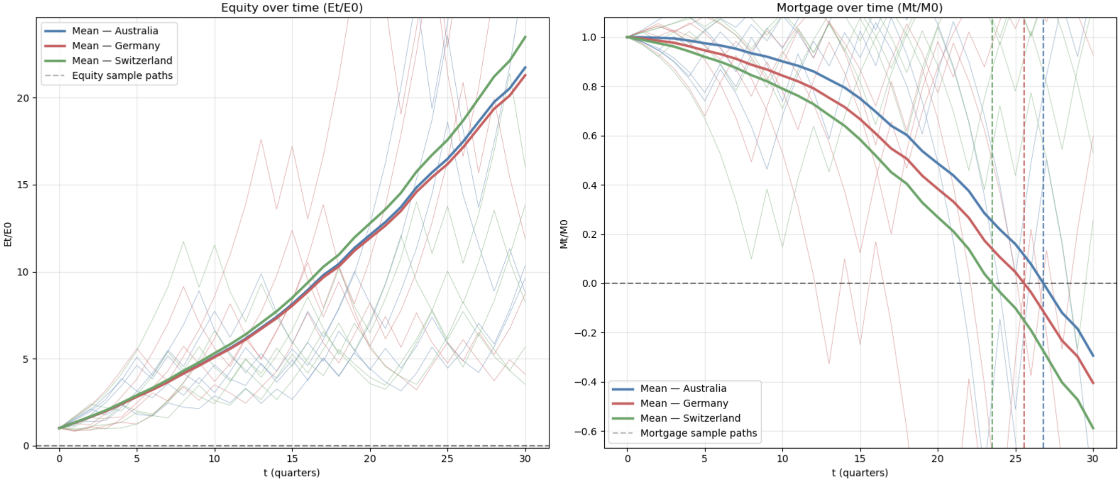
Results 2.1: Self-Owned — Phase Diagram

The y axis uses the country specific s band set in and the red and green marks show the default and success scenarios for the Monte Carlo simulation

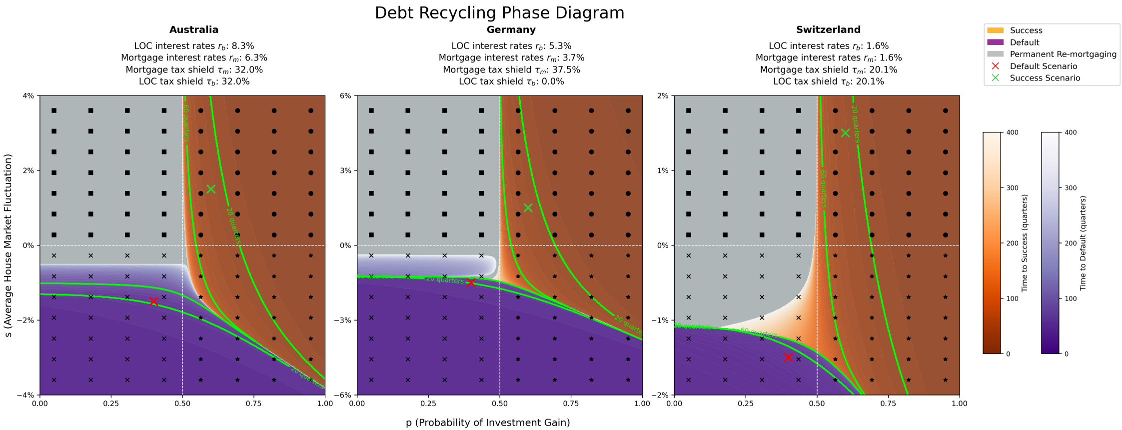


Results 2.2: Self-Owned — Monte Carlo

Success scenario uses $p = 0.6$ and $s = 1.5\%$

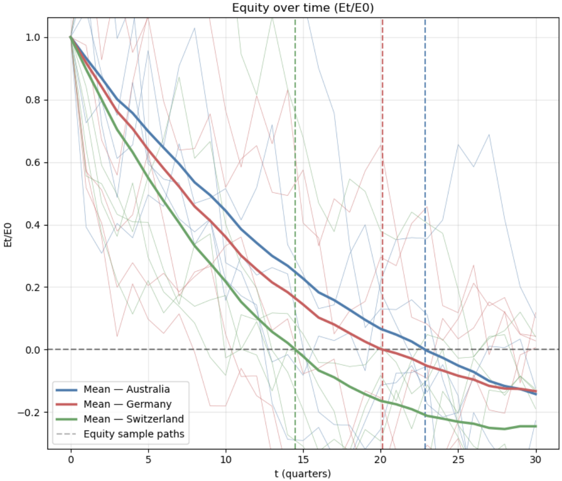


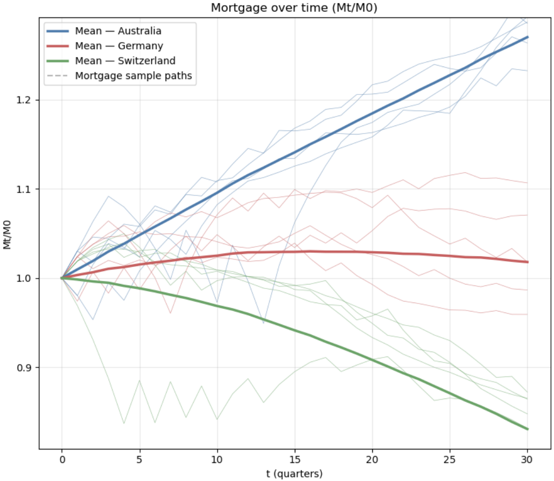
Same marks for success and default and the y axis uses the country specific s band



Results 2.4: Rental — Monte Carlo

Default scenario uses $p = 0.4$ and $s = -1.5\%$





Discussion — Calibration

- Self-owned Switzerland most favourable due to low rates and a mortgage shield Germany next Australia last
- Rental shields widen success for all three countries ordering still driven by interest rates and tax shields
- Interesting rental default case with $p = 0.4$ and $s = -1.5\%$ Switzerland can hit $E \leq 0$ first because very low r_m limits the equity boost from $\tau_m r_m M$ as Australia and Germany both have a mortgage shield and higher interest rates.
- Phase diagrams and Monte Carlo tell the same story boundaries shift as expected and time differences match the geometry

Conclusion, Future Work and Main References

Key Takeaways

- Rates shrink success regions and slow mortgage payoff, tax shields offset part of this
- Calibrated results line up with country rules and explain where debt recycling helps

Future Work

- Add an explicit refinance stop and treat success, default, and refinance as separate exits with times
- Use richer tax logic: imputed rent in Switzerland, a small deduction floor in Germany, and canton caps in Switzerland
- Model dependence between housing returns and the risky asset

Main References

- Aufiero et al. 2025 Base model for debt recycling
- Australian Taxation Office guidance on interest deductibility
- German EStG interest deduction rules
- Swiss federal tax code on imputed rent and deductions